Course Overview

This course investigates the fundamental issues addressed by scientists studying tropical rain forests, including what gave rise to their remarkable biodiversity, the drivers and consequences of deforestation, why people are fascinated by rain forests, cultural and ecological stereotypes about the tropics, and different approaches to forest conservation and sustainability. ***By the end of the course students will be able to:***

- Recognize and describe stereotypes about rain forests & their residents
- Analyze rain forest tropes in art, literature, & popular culture
- Discuss & evaluate hypotheses for the origins and maintenance of tropical biodiversity
- Explain & compare the history and ecological influence of humans in rain forests
- Review contemporary threats to rain forests
- Analyze and visualize data on deforestation
- Review and contrast strategies for rain forest conservation & restoration
- Identify rain forests in their daily lives & set personal goals for advancing their conservation
- Produce materials for communicating about tropical biology & conservation to family and peers

When & Where

When: Tuesdays 11:45 AM-12:35 PM (Period 5) & Thursdays 11:45 AM-1:40 PM (Periods 5-6)
Where: Larsen Hall 0310 (both days)

Instructor & TA

Instructor: Dr. Emilio M. Bruna (email: embruna@ufl.edu, phone: (352) 846-0634)

Credits & Prerequisites

Credits: 3
Prerequisites: None
Quest Program: Quest 2
GenEd Designation: International

A minimum grade of C is required for Quest and General Education credit. Courses intended to satisfy Quest and General Education requirements cannot be taken S-U.

Minors & Certificates

This course counts towards the Minor in Latin American Studies. See https://www.latam.ufl.edu/academics/undergraduate-programs for more information.

University Policies and Resources

Information about grading policies, support for students with disabilities, course evaluations, the Honor Code, and other course policies and campus resources can be found on the Syllabus Policies page.

Information about university attendance policies, including policies regarding religious holidays, illness, and the twelve-day rule, can be found in the Undergraduate Catalog.

Required Materials

All materials, including readings and videos, will be made available on the course Canvas page.

Materials and Supplies Fees: None.

Textbooks to purchase: None.

> **!** Important
>
> *Students should sign up for free online access to the New York Times, Wall Street Journal, and The Economist by following the instructions at this UF Libraries Website.*

Office Hours

When and Where

- **Instructor:** Wednesday 1:30-3:30 (in-person in the TEC Lab (see below) and via Zoom. If you want to guarantee a specific time slot you can do so by signing up here: https://embruna.youcanbook.me/. If you can't make it during the regularly scheduled office hours, you can send a message via Canvas to set up an appointment.
- **Teaching Assistant:** Tuesday 4:00-5:30 pm (in-person in the TEC Lab & online) or by appointment.
- **In person location:** The Tropical Ecology & Conservation Lab is located at 711 Newell Drive next to Rawlings Hall (between the bus stop and the parking garage). The location and directions are on Google Maps.
- **Online:** To join Office Hours virtually use the Zoom Conferences link on the main menu of the course Canvas page. We are online the entire session.

> Can you give me **one good reason** why I should go to Office Hours?
>
> I can give you **ten**.
>
> 1. To introduce yourself.
> 2. Get clarification on assignments.
> 3. Argue about a topic that came up in class.
> 4. Grab a (free) tea, coffee, or espresso in our lab kitchen.
> 5. Make sure you understood the key points from a class session.
> 6. Ask for feedback on ideas for course projects.
> 7. Get advice on successfully navigating academic life at UF.
> 8. Discuss how to gain experience for your post-graduation goals.
> 9. Request help arranging a study group.
> 10. You don't need a good reason…just come on by.

Table 0: Assignments for WIS 2323 and their point value.

Assignment	Points	Percent
In-class activities	550	55%
Analytic Essay	200	20%
Reflective Essay	150	15%
Movie Reviews	100	10%

Grades & Attendance

Learning in our class is achieved with an diverse array of methods ranging from data analysis to essays to projects. In most class sessions you will also be working with small groups of students to complete an in-class assignment that reinforces the major themes of the day's topic. In keeping with the philosophy of the Quest program, this course also has *Experiential Learning and Self-Reflection Components*.

Important note regarding class discussions & group work.

We will explore some challenging, important problems and increase our understandings of different perspectives and approaches for addressing them. These conversations may not always be easy; we sometimes will make mistakes in both how we communicate our perspective and what we hear other say. There may be times when we need patience, courage, imagination, and of course mutual respect to engage our texts, classmates, instructors, guests, and our own ideas and experiences. *Disrespectful or disruptive behavior will not be tolerated.* And always remember that as scholars we rely on critical thinking, data, prior scholarship, and expert opinion when interrogating assigned readings and discussing course content with classmates and instructors.

Assignments (1000 pts total)

Grading

In-class Assignments are due by the following class session. Late assignments will lose 10 pts.

Regrades: Requests for re-evaluation of assignments must be accompanied by an explanation for why you think you deserve additional credit and the number of additional points you think you deserve. The deadline for submission is one week after the work was returned.

Grade Assignment (based on % of total points earned): A = 94–100%, A- = 90–93%, B+ = 87–89%, B = 84–86%, B- = 80–83%, C+ = 77–79%, C = 74–76%, C- = 70–73%, D+ = 67–69%, D = 64–66%, D- = 60–63%, E<60

Grade Points: For information on how UF assigns grade points, visit: https://catalog.ufl.edu/UGRD/academic-regulations/grades-grading-policies/

Attendance

Though attendance is not required, many of the sessions we will be completing activities in class that count towards your grade. Some these can be completed independently, but by doing them in class you will benefit from working collaboratively with the other students. **Not all in-class activities can be completed independently.** That is why only a subset of the in-class assignments count towards your grade. We also offer some opportunities for extra credit. ***If you need to miss class for any reason, please let me know as soon as possible***. We will try to make arrangements for you to complete any assignments and go over any material you will be missing.

Participation

Consistent informed, thoughtful, and considerate class participation is encouraged (and in some cases required). *If you have personal issues that prohibit you from joining freely in class discussion (e.g., shyness, language barriers, medical condition): no problem.* let us know and we will discuss alternative modes of participation.

Topic Calendar & Key Dates

FAQs

(1) **How do I contact the Instructor or TA?** Send a message in CANVAS. That way both the TA and instructor will see it and you will get a response more quickly (emails sent to us directly tend to get buried under other messages).

(2) **How will you send class announcements?** CANVAS. Check the course page for announcements and be sure you are receiving Canvas emails and updates.

(3) **What work should we complete BEFORE class?** Read, watch, listen to, or review all materials assigned for the session. This material sets the stage for in-class activities.

(4) **What will we do DURING class?** In-class exercises that reinforce key concepts, discussions of the assigned readings, and let you practice skills in other assignments. Some are completed individually, while others require working in groups or pairs. Each activity will have instructions and a rubric; most are designed to be finished in class.

(5) **When is the "in-class" work due?** Deadlines are in Canvas, but typically the start of the next class session.

(6) **What if I miss class?** Attendance is not required, but in many of the sessions we will be completing activities in class that count towards your grade. Some of these can be completed independently, but by doing them in class you will benefit from working collaboratively with the other students. However, some of the in-class activities can not be completed on your own. That's why only a subset of the in-class assignments count towards your grade and we offer extra credit.

(7) **I have to miss class on a certain date. What should I do?** Let us know as soon as possible so we can make arrangements for you to review material you will miss and complete assignments.")

(8) **Class discussions are difficult for me. Will this affect my grade?** No! If there are issues that make engaging in discussions difficult (e.g., shyness, language barriers, a medical condition), let us know and we will find alternative modes of participation.

(9) **I unexpectedly have no child care today...what can I do?** UF does not have a policy on children in the classroom, so here is mine: I never want students to feel they have to choose between their responsibilities as parents and their education. Bringing your child to class because of a gap in child care is totally fine (and that includes babies). If you do so, please try and sit close to the exit so that you can more easily step outside if you need to care for them and so other students can continue learning.

Table 0: Course Topics and Key Dates

WE	DAT	TOPIC	REA ING
Wk 1	20-Aug	Course Welcome and Introduction	
		WHY ARE WE FASCINATED BY TROPICAL RAIN FORESTS?	
Wk 2	25-Aug	Historical Narratives	1
	27-Aug	Historical Narratives, cont.	
Wk 3	1-Sep	Rain forests in Art & Literature	2
	3-Sep	Rain forests in Pop Culture	3
		THE ECOLOGY & EVOLUTION OF TROPICAL RAIN FORESTS	
Wk 4	8-Sep	What is a rain forest?	4,5
	10-Sep	Where is a rain forest?	
Wk 5	15-Sep	Patterns of biodiversity	6,7
	17-Sep	Patterns of biodiversity, cont.	8
Wk 6	22-Sep	Origins of tropical biodiversity	
	24-Sep	Maintenance of tropical biodiversity	
Wk 7	29-Sep	Humans in rain forests	9,10
	1-Oct	Humans in rain forests, cont.	
Wk 8	6-Oct	Paradox of Luxuriance & (Bio)Narratives Revisited	11,12
	8-Oct	JUNGLE FILM FESTIVAL	
		THE DRIVERS AND IMPACTS OF DEFORESTATION	
Wk 9	13-Oct	How much rain forest is there?	13

	15-	How much rain forest have we lost?	14

Assigned Reading & Viewing

> ❗ Important
>
> Topics & readings are subject to change based on student interests and current events; changes will be announced via Canvas. *Please read or watch all assigned material before class.*

Week 1

Thursday: Course Introduction

- *No readings for the first day of class.*

Week 2

Tuesday: Historical Narratives

Goal: *To introduce and discuss how historical narratives by the first Europeans to visit the tropics have shaped contemporary perceptions of tropical rain forests and the colonial roots of tropical biology*

(10) Excerpts from Historical Narratives (read online, 6 pages)

Thursday: Historical Narratives (cont.)

- *No additional readings.*

Week 3

Tuesday: Rain Forest Imagery in Art & Literature

Goal: *To see how the different depictions of the tropics and tropical biodiversity from early European explorers are reflected in art and literature.*

(11) Excerpts from Literature/Poetry (link, 5 pages)

(12) Images of artwork (link,5 pages)

Thursday: The Rain Forest in Pop Culture

Goal: *To compare the use and presentation of rain forest images by the private sector and in different forms of popular culture, including the film and music industries, and to evaluate how these depictions influence perceptions of tropical countries & people.*

(13) Jolly, Priscilla. (2021). 'Godzilla vs. Kong': Monster movies evoke adventure but also 'dangers' of tropics. *The Conversation.* (link; 4 pages)

Week 4

Tuesday: What is a Rain Forest?

Goal: *To learn the different ways biologists define the "tropics" and how the structure and dynamics of tropical rain forests differ from those of forests in other parts of the world.*

(14) Why Does Earth Have Deserts? (link to video; 2 min)

(15) BBC Planet Earth: Jungles. (2006). link to video; requires login with Gatorlink, 51 min).

Thursday: Where is a Rain Forest?

Goal: *To understand the geological history of tropical rain forests, how climate, fire, and geological history drive the tipping point between forests and savannas, and how this biogeographic, geological, and climatic history shaped the evolution of tropical plants and animals.*

- No readings for today's class

Week 5

Tuesday: Patterns of Biodiversity

Goal: *To observe and catalog the diversity of plant and animal life forms that can be found in rain forests, to quantify the local patterns of species richness and abundance in a tropical forest, and compare these patterns with those in the temperate zone.*

(16) CBS News. (2023). Inside the Smithsonian's tropical forest lab. (watch online, 5:35 min.)

(17) Butler, Rhett. (2015). How many tree species are found in the world's rainforests? *Mongabay* (read online, 5 min.)

Thursday: Patterns of Biodiversity, continued

Goal: *To understand global patterns of species richness and how these vary from the tropics to the temperate zone.*

(18) Matthew Earl Boone and Mathieu Basille. Using iNaturalist to contribute your nature observations to science. UF EDIS Document WEC413 (read online, 4 pages) and then familiarize yourself with `iNaturalist` at https://www.inaturalist.org/

Week 6

Tuesday: The Origins of Tropical Biodiversity

Goal: *To review hypothesized mechanisms for the origins of tropical diversity and the role of interspecific interactions in the (co)evolution and diversification of tropical biodiversity*

- No readings for today's class

Thursday: The maintenance of tropical biodiversity

Goal: *To review the biotic and abiotic mechanisms in tropical rain forests that permit the coexistence of so many species*

- No readings for today's class

Week 7

Tuesday: Humans in Rain Forests

Goal: *To understand the history of human occupation of rain forests including the contemporary demographic transition from rural to urban occupation; to review the different ways in which humans have historically modified rain forests and how this has shaped current rain forest biodiversity.*

(19) Allington, Kristine, Michael Amundson, Linithd Aparicio, Caitlin Saks (Producers), and G. Townsley (Director). (2023). Ancient Builders of the Amazon. Public Broadcasting Service. (watch online, 53 min.)

(20) Hunt, Chris and Premathilake, Rathnasiri. (2018). Prehistoric people started to spread domesticated bananas across the world 6,000 years ago. *The Conversation.* (read online, 3 pages).

Thursday: Humans in Rain Forests, continued

- No readings for today's class

Week 8

Tuesday: Paradox of Luxuriance

Goal: *To understand how such a productive biome can be built on such low-quality soils, and explore the implications of this "Paradox of Luxuriance".*

(21) Anthony Bourdain's Parts Unknown: Congo (S1E8) (*watch online free on* XUMO PLAY *or* PLEX, 50 min).

(22) Brooks, David. (2010). The Messiah Complex. *The New York Times.* (read online, 1 page)

Thursday: Jungle Film Festival

- No readings for today's class

Week 9

Tuesday: How much tropical rain forest is there?

Goal: *To learn how forest cover is defined and estimated and how it varies globally.*

(23) Louis Lucero. (2013). New Interactive Tool Helps Track Earth's Forests. *New York Times.* (read online, 10 min read).

Thursday: How much tropical rain forest have we lost?

Goal: *To use forest cover data to estimate rates of tropical forest loss over time.*

(24) Serkez, Yaryna. (2020). Every Place Under Threat. *New York Times.* (read online, 10 min read).

Week 10

Tuesday: Drivers & Ecological Consequences of Deforestation

Goal: *Learn (a) how deforestation and other human activities alter the structure & functioning of rain forests, and (b) compare how these drivers differ between the African, American, and Asian tropics.*

(25) Devouring the Rain Forest. *Washington Post.* (read online, 20 min read).

(26) Searcey, Dionne with photographs by Ashley Gilbertson. (2022). Raft by Raft, a Rainforest Loses Its Trees *New York Times.* (read online, 10 min read).

(27) Andreoni, Manuela, Blacki Migliozzi, Pablo Robles and Denise Lu with photographs by Victor Moriyama. (2022). The Illegal Airstrips Bringing Toxic Mining to Brazil's Indigenous Land. *New York Times.* (read online, 15 min read).

Thursday: Drivers & Ecological Consequences of Deforestation, continued

Goal: *Continue learning (a) how deforestation and other human activities alter the structure & functioning of rain forests, and (b) compare how these drivers differ between the African, American, and Asian tropics.*

(28) Mason, Margie & McDowell, Robin. (2020). Palm oil labor abuses linked to world's top brands, banks. *Associated Press.* (read online, 10 min read).

(29) Lustgarten, Abrahm. (2018). Palm Oil Was Supposed to Help Save the Planet. Instead It Unleashed a Catastrophe. *New York Times.* (read online, 20 min.)

Week 11

Tuesday: Tropical Forests & Global Climate

Goal: *Climate change and Tropical Forests To understand the relationship between tropical forests, deforestation, and the global climate cycle.*

(30) BBC News: Amazon rainforest: 'Once it's gone, it's gone forever' - interview with Erika Berenguer. (link to video, 3 min).

(31) Pearce, Fred. (2018). Rivers in the Sky: How Deforestation Is Affecting Global Water Cycles. *Yale360*. (read online, 10 min read).

Thursday: Tropical Forests & Climate Change

(32) Henry Knight Lozano (2023) California and Florida grew quickly on the promise of perfect climates in the 1900s – today, they lead the country in climate change risks. *The Conversation*. Link

(33) learn about the EN-ROADS simulator we will be using in class at this (link to website, 15 min read). You can even start experimenting with it here: (link to site).

Week 12

Tuesday: International Conservation Frameworks

Goal: *To learn about the major local, national, and multi-national approaches to reducing deforestation by comparing their efficacy and socioeconomic impacts. REDD and Payment for Ecosystem Services.*

(34) Dasgupta, Shreya. 2017 Does forest certification really work? *Mongabay* (read online 15 min.)

(35) Gaworecki, Mike. (2017). Cash for conservation: Do payments for ecosystem services work? *Mongabay* (read online 15 min.)

(36) Gaworecki, Mike. (2021). How effective is the EU's marquee policy to reduce the illegal timber trade? *Mongabay* (read online 15 min.)

Thursday: Trade in tropical commodities & DURIAN FEST

Goal: *To learn about the global market in tropical fruit crops and the economic impact of tropical fruit production in Florida.*

(37) Ryan, John Charles. (2018). Why the stinky durian really is the 'king of all fruits'/ *The Conversation*. (read online, 10 min).

(38) Weintraub, Karen. (2019). They're Smelly and Spiky, and They Need Bats to Pollinate Them. *New York Times*. (read online, 5 min read).

(39) Wharton, Rachel. (2020). How the Tip of Florida Became a Tropical-Fruit Paradise. Atlas Obscura. (read online, 5 min read).

(40) NPR's Throughline Podcast (2022). There Will Be Bananas. [listen online, 56 min].

Week 13

Tuesday: Consumer choices & rain forest conservation

Goal: *To understand how global production and chains and consumer demand in Europe and North America influence patterns of deforestation in tropical countries.*

(41) Lawal,Shola. (2020). Our Endless Appetite For Chocolate Has Bitter Environmental Consequences. *Huffington Post*. (read online, 10 min read).

(42) Mufson, Steven and Georges, Salwan. (2019). The trouble with chocolate. *Washington Post* (read online to see the amazing maps, pictures, and data visualizations).

(43) Williams, Wyatt. (2021). How Your Cup of Coffee Is Clearing the Jungle. *New York Times*. (read online, or listen to the audio version with the link at top of the article).

(44) Carodenuto, Sophia. (2021). Chocolate fix: How the cocoa industry could end deforestation in West Africa. *The Conversation*. (read online, 10 min read).

(45) Mary Gagen. 2024. The world's business and finance sectors can do much more to reverse deforestation – here's the data to prove it. *The Conversation*. (read online, 10 min).

Thursday: Community based conservation

Goal: *To learn about how local communities are engaged in rain forest conservation and sustainable development efforts in the tropics and beyond.*

(46) Katrina Kosec et al. (2025) Forest loss in Malawi: how having women at the table affected debates and decisions about solutions. *The Conversation*. (read online, 10 min read).

(47) *BBC World Service* (2020). Don't underestimate what one person can do on their own: The man who grew his own rainforest. (link to video, 5 min long).

(48) Dasgupta, Shreya. (2017). Does community-based forest management work in the tropics? *Mongabay* (read online, 15 min read).

(49) Maclean, Ruth with additional reporting by Caleb Kabanda and photography by Nanna Heitmann. (2022). What do the protectors of Congo's peatlands get in return?" *New York Times*. (read online, 10 min read).

Week 14

- *No readings (Thanksgiving Holiday)*

Week 15

Tuesday: Protected areas

Goal: *To learn about different global categories of protected areas, the importance of protected areas in the tropics for conserving forest, and how the threats to protected areas vary regionally and globally.*

(50) Kimbrough, L. (2021). How settlers, scientists, and a women-led industry saved Brazil's rarest primate". *Mongabay* (read online, 10 min read).

(51) Dasgupta, Shreya (with research by Annika Schlemm & Zuzana Burivalova). (2017). Do protected areas work in the tropics?" Mongabay. (read online, 25 min read).

(52) Matti Barthel et al. (2025) DRC's plan for the world's largest tropical forest reserve would be good for the planet: can it succeed? *The Conversation*. (read online, 10 min read).

Thursday: Forest restoration & Regeneration

Goal: *To learn the difference between passive regeneration and active restoration and assess evidence for whether they can be used to reverse the effects of deforestation.*

(53) Gaworecki, Mike. (2021). Is planting trees as good for the Earth as everyone says? *Mongabay* (read online 15 min.)

(54) Medici, Patricia. (2015). The coolest animal you know nothing about...and how we can save it. *TED Fellows Talk*. (link to video, 11:20 min long).

(55) Robin Chazdon et al. (2021). Tropical forests can recover surprisingly quickly on deforested lands – and letting them regrow naturally is an effective and low-cost way to slow climate change. *The Conversation*. (read online, 10 min.).

Week 16

Tuesday: Tropical rain forests & Global Health

Goal: *To learn about the relationship between deforestation and the emergence and spread of tropical diseases like Zika and Malaria from the tropics to other regions of the globe.*

(56) Vittor, Amy, Gabriel Zorello Laporta, and Maria Anice Mureb Sallum. (2020). How deforestation helps deadly viruses jump from animals to humans. *The Conversation*. (read online, 15 min read).

(57) UF EPI. (2024) Florida's mosquitoes can make you sick: Here's how to protect yourself. read online, 10 min. read)

(58) Kuchipudi, Suresh V. (2020). Why so many epidemics originate in Asia and Africa – and why we can expect more. *The Conversation*. (read online, 10 min).

Student Learning Objectives

B = Biological Sciences, Q2 = Quest 2

CONTENT

Students demonstrate competence in the terminology, concepts, theories, and methodologies used within the discipline(s)

(1) Describe the evolutionary and ecological factors underlying the distribution of biodiversity in tropical rain forests. *Assessments:* In-class activities. (B).

(2) Distinguish (a) the different ways humans use and alter rain forests, and (b) the social, economic, and biological consequences of these activities. *Assessments:* In-class activities, Reflective Essay. (B, Q2)

(3) Explain how genetics, remote sensing, computational tools, and other scientific developments have advanced research on the ecology and evolution of rain forest biota. *Assessments:* In-class activities, Analytic Essay. (B).

(4) Examine the historical factors that have shaped cultural perceptions of rain forests. *Assessments:* In-class activities, Movie Reviews. (Q2).

(5) Analyze how narratives about rain forests are reflected in different types of contemporary cultural expression. *Assessments:* In-class activities, Movie Reviews. (Q2).

(6) Distinguish between the primary drivers of forest loss and how they vary geographically. *Assessments:* In-class activities, Analytic Essay. (B, Q2)

(7) Critique different proposed mechanisms for rain forest conservation. *Assessments:* In-class activities, Analytic Essay. (B, Q2).

CRITICAL THINKING

Students carefully and logically analyze information from multiple perspectives and develop reasoned solutions to problems within the discipline(s).

(1) Test hypotheses regarding trends in deforestation and how forest regeneration varies geographically with quantitative data on forest cover and use news stories, primary literature, and other sources to propose mechanisms responsible for the patterns observed. *Assessments:* In-class activities, Analytic Essay (B, Q2).

(2) Analyze qualitative data on rural-urban migration and human demographic shifts in tropical countries and assess the potential implications of the results for conservation and broader societal issues. *Assessments:* In-class activities, Analytic Essay (B, Q2).

(3) Explain the relationship between deforestation and the global climate cycle. Compare alternative policy pathways for reducing CO2 emissions using the En-ROADS global climate simulator. *Assessments:* In-class activities, Reflective Essay (B, Q2).

COMMUNICATION

Students communicate knowledge, ideas and reasoning clearly and effectively in written and oral forms appropriate to the discipline(s)

(1) Generate graphical analyses of quantitative data used to study rain forest biology and conservation (Q2, B). *Assessments:* In-class activities, Analytic Essay.

(2) Compose summaries of research on multidisciplinary questions relevant to tropical forest biology and conservation based on logical arguments (Q2, B). *Assessments:* In-class activities, Analytic Essay.

(3) Develop audience-specific content and materials with which to communicate an important issue related to tropical forest biology and conservation. *Assessments:* In-class activities, Reflective Essay, Movie Reviews, Analytic Essay, Final Project (Q2).

CONNECTION

Students connect course content with meaningful critical reflection on their intellectual, personal, and professional development at UF and beyond.

(1) Examine the cultural, economic, and historical experiences of people in rain forest countries, how this compares with our preconceived notions of the same, and the consequences of this disparity for our scientific understanding of rain forests and their conservation. *Assessments:* In-class activities, Movie Reviews, Final Project (Q2).

(2) Assess the feedbacks between (a) the actions of individuals, governments, and the private sector, (b) global economic, social, and political conditions, and (c) the status rain forests and the global climate cycle. *Assessments:* In-class activities, Reflective Essay, Analytic Essay (Q2).

(3) Appraise the ubiquity of tropical forest products in daily life, the role and impact of the global commodity chains that make this possible, and the consequences of consumer behavior for forest conservation and socioeconomic sustainability. *Assessments:* In-class activities, Final Project (Q2).

(4) Analyze the local, regional, and global ecosystem services provided by tropical rain forests, how these vary geographically, and some of the cultural and ecological factors responsible for these differences. *Assessments:* In-class activities, Reflective Essay, Analytic Essay, Final Project (Q2)